Every Version of Me: Engineering Design Document

Sep 27, 2026

1. Overview

Every Version of Me is a 12-room narrative pixel game shipped as one 189,678-byte HTML page that opens in the browser on her MacBook. Apart from two web fonts, it has no dependencies, no build server and no assets outside the file: every sprite, tile, song and sound effect is generated at runtime from code.

Metric	Value
Source file	2,397 lines, 189,678 bytes
Runtime script	2,141 lines, 174 named functions
Fonts	Pixelify Sans and IM Fell English, loaded from Google Fonts
Logical framebuffer	240 × 160 px, 16 px tiles, 15 × 10 tiles per room
Content	12 rooms, 12 letters, 16 hidden hearts, 11 badges, 11 outfits, 13 stickers
Challenges	12 room tasks: 8 overlay minigames, 4 in-world tasks
Audio assets	0 files; 11 instruments and 8 arrangements synthesised live
Image assets	0 files; 40+ sprites compiled from string grids at boot

Hard constraints

- One HTML page, opened in a browser with nothing to install. The primary target is her MacBook.
- Works in Chrome, Edge, Firefox and Safari on desktop, with touch fallback on phones and tablets.

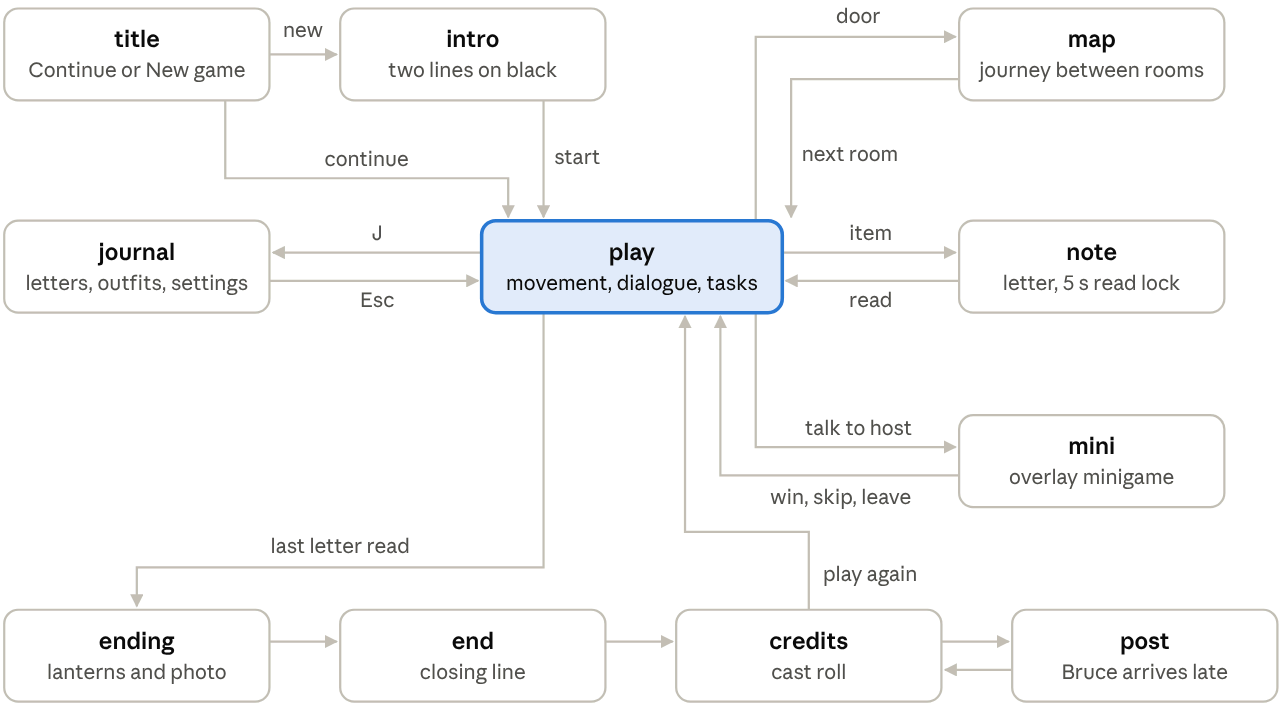
- Every personal detail (letters, hearts, names, colours, dialogue) is editable in one `CONFIG` object without touching engine code.
- The player can never be blocked from a letter: every challenge has a guaranteed path to completion.

Design goals, in priority order: emotional delivery of the letters, zero-friction launch, charm density per room, replay value, and maintainability by a non-engineer.

2. System architecture

The game is a single-threaded, immediate-mode engine: one `requestAnimationFrame` loop calls `update(dt)` then `draw()`, and a global mode flag `G.mode` decides who owns input and the frame.

Every screen is a mode, and play is the hub the others return to



runtime modes · 11 states, play is the hub

Dialogue is deliberately not a mode. It is a modal overlay (`dlg`) that pauses movement inside `play`, so a cat can speak during a task without a state transition.

Frame loop contract

1. `dt` is clamped to 50 ms, so a backgrounded tab resumes without teleporting the player.
2. Timers, pet animations, emotes and particles tick first, in every mode, so the world stays alive behind overlays.
3. The first mode handler that claims the frame returns early: note, credits, mini, post, fade, dialogue, map, then play.
4. `draw()` renders the room, sorts entities by y, draws particles, then tint, vignette, minigame overlay and fade, in that order.

Module map. The script is layered so each layer only calls downward: CONFIG (content) → sprite compiler and tile rasteriser → world model and rooms → dialogue, notes and tasks → minigames → audio engine → persistence → shell (title, journal, credits, layout). Later layers extend earlier ones by wrapping function bindings (`drawObj` , `drawWall`) rather than editing them, which kept 14 feature passes additive.

3. Rendering pipeline

All art is compiled at boot from text: 40+ sprites are declared as string grids with a palette map, rasterised once into offscreen canvases, and blitted with `drawImage` every frame.

Sprite compiler (`makeSprite`)

- Input is a list of equal-width strings, one character per pixel, `.` for transparent, plus a `{char: colour}` palette.
- The output canvas is 2 px larger than the grid. A 4-neighbour dilation pass paints every empty pixel that touches a filled one with the outline colour.
- Black cats (Salem, Shadow) and Poe take a lavender rim instead of a dark outline, so a near-black sprite still reads on a dark floor.
- Variants are derived, not drawn: `flipped()` mirrors via a canvas transform, `rotSprite()` rotates arrows by quarter turns, and Shadow's sleeping face is his awake grid with the eye character swapped by regex.

Tile rasteriser. Each frame redraws 150 tiles immediately; there is no cached tilemap, so animated water, flickering candles and blooming roses cost nothing extra. Organic detail (grass tufts, pebbles, stars, city windows) comes from `hh(x, y, seed)` , an integer hash built on `Math.imul` , which gives stable per-tile noise with zero stored textures.

```
hh(x,y,s)=\operatorname{fmix}\!\left(x\cdot 374761393 + y\cdot 668265263 +
s\cdot 1442695041\right) / 2^{32}
```

Draw order, back to front

1. Sky band (night gradient, stars, moon or skyline) for outdoor rooms.
2. Floor, then the object layer for each tile, from 22 tile characters.
3. Entities, painter-sorted by tile row: items, NPCs, the follower at y minus 0.02, the player at y plus 0.01.
4. Particles (11 kinds, from lantern and firefly to rain and "zz"), then tint, radial vignette, minigame overlay and fade.

Scaling. The 240 × 160 canvas is upscaled by CSS with `image-rendering: pixelated`. The scale snaps to quarter steps once it reaches 2×, and the same factor is exported as a CSS unit (`--u`), so HTML overlays (dialogue, bubbles, HUD) scale in lockstep with the pixel art.

HTML over canvas. Text never lives in the canvas. Dialogue, speech bubbles, notes and the journal are DOM elements, which keeps type crisp at any scale and lets letters use a real serif. Bubbles are re-anchored every frame to their speaker, clamped to the stage edge, and flipped below the speaker when they would clip the top.

Keepsake photo. At 5.2 s into the ending, the framebuffer is redrawn at 4× nearest-neighbour into a 1,032 × 836 polaroid with a serif caption and the play date, then encoded as PNG.

4. World model and data schema

The world is 12 rooms of 15 × 10 tiles, each declared as ten 15-character strings plus a small record of host, task, items and mood. Rooms are rebuilt fresh on every new game, so no stale state survives a replay.

#	Room	Host	Task	Letter item	Music	Ambience	
0	Ellie's Room	George	search	polaroid	room	none	
1	Moonlit Garden	Salem	fireflies	black rose	garden	crickets	
2	Record Shop	Peaches	rhythm	record	shop, metal in game	vinyl	
3	Royal Kitchen	George	recipe	recipe card	room	candle	
4	Candlelit Library	Shadow	battle	book	library	candle	
5	Quiet Pond	Salem	fishing	bottle	garden	crickets	

#	Room	Host	Task	Letter item	Music	Ambience	
6	Rainy Rooftop	Poe	catch	paper crane	rain	rain	
7	Night Market	Pickle	bop	fortune cookie	garden	crickets	
8	Frozen Cave	Shadow	ice	snow globe	library	wind	
9	Observatory	Poe	stars	star map	rain	wind	
10	Late-Night Café	Peaches	café	boarding pass	shop	vinyl	
11	Balcony	all four cats	lanterns	last letter	balcony	crickets	

Tile grammar. 33 single-character tiles, split by one rule: a tile is solid if it is in the `SOLID` set, and a door is solid until its room's task is resolved.

- Walkable: `.` floor, `f` flowers, `d` path, `r` rug, `q` puddle, `n` dock, `K` blanket, `~` ice (walkable but slippery).
- Solid scenery: walls, trees, bushes, water, rocks, sky, railings, and 15 furniture tiles from beds and bookshelves to the fridge, telescope and kettlebell.
- Behaviour tiles: `D` door (locked, then glowing), `m` letter pedestal, `L` lantern (lit or unlit per task state), `V` record player (jukebox).

Entities

- NPC: `{id, x, y, talk(), room, petT, saidAfter}`. The first NPC in a room is its host by convention; the speech-bubble, badge and pet systems all rely on that.
- Item: `{type, x, y, key?, hidden?, note?, trait?, taken}`. A `key` item carries the room's letter index; a heart carries its fixed trait index.
- Follower: `{id, x, y, tx, ty, hidden}`. It steps into the tile the player just left, and hides itself in any room where its cat is an NPC.

Task lifecycle. Every room task moves through four one-way flags, and each transition has exactly one owner:

1. `started`, set when the host's intro dialogue ends.
2. `complete`, set by the minigame or world task.
3. `done`, set by the host's closing dialogue, which awards the badge and reveals the letter item.
4. Door open, set when the letter is closed after its read lock.

CONFIG surface. All content lives in one object with 43 top-level keys, including 77 dialogue scripts (115 lines), 12 letters, 16 hearts, 13 stickers and 11 outfits. The engine reads names through accessors (`SPEAKERS.george.name()`), so renaming a cat in one place renames it everywhere, credits included.

5. Narrative systems

Every word the player reads flows through one small dialogue VM and one template function, so tone, timing and personalisation are controlled in a single place.

Dialogue VM (`say`)

- A script is an array of line objects: `{who, text, item?, do?}` . `say(lines, onDone)` enqueues them and returns immediately.
- Lines type out at 40, 95 or effectively infinite characters per second (the text speed setting). A key press first completes the line, then advances.
- `do` fires a side effect the moment its line appears: `reveal` uncovers the letter item, `open` unlocks the door, `bloom` opens the garden roses. Story beats and world changes can never drift out of sync.
- `onDone` continuations chain flows: host intro, then `startTask` , then minigame, then `finishTask` , then badge, outfit and closing dialogue.
- Each speaker has a voice: a pitched blip per three characters while typing, plus a synthesised meow, caw or ribbit when a new speaker starts.

Template engine (`fmt`). One regex pass replaces `{key}` tokens from a live map: `{name}` resolves to the configured name, `{n}` to the current task counter. Unknown tokens pass through untouched, so a typo shows up on screen rather than crashing.

Letters with an enforced read time. Opening a letter switches to `note` mode, ducks the music to a solo piano arrangement, and starts a 5 s lock drawn as an SVG ring whose `stroke-dashoffset` drains linearly:

```
\text{offset}(t)=C\left(1-\frac{\max(0,\,5-t)}{5}\right),\qquad C=2\pi\cdot 18\approx 113.1
```

The continue button stays disabled, a clock tick sounds each second, and pressing Enter early shakes the timer instead of closing. Rereading from the journal skips the lock.

Hidden hearts. Each of the 16 hearts is bound to a fixed trait index, so the eyes heart is always the last one, on the balcony. Hearts are invisible from afar and fade in as the player approaches, with opacity driven by tile distance d:

```
\alpha(d)=\operatorname{clamp}\!\left(\frac{4.2-d}{2},\,0,\,1\right)
```

A faint sparkle is always emitted, so a careful player can find them without the game pointing.

Reactive personality layer

- Outfit reactions: a host comments once per cat per worn item per run, before its normal dialogue.
- Replay awareness: when a previous run was completed, each host's first intro line is swapped for an "Again" variant.
- Salem's chatter: every 22 to 38 s while following, drawn from a shuffled bag so no line repeats until all have played. Each room's own line fires once, with a 70% chance per turn.
- Speech bubbles: purely cosmetic DOM elements that never take input focus, so they cannot interrupt play.

6. Minigame framework

All eight overlay minigames implement one duck-typed interface and share one host, so adding a game means writing a factory, never touching the loop.

The `MINI` contract

- Required: `title`, `hint`, `hintTouch`, `init()`, `update(dt)`, `draw()`, `onEnd(ok, score)`.
- Optional: `onDir(dir)` (edge-triggered from keys or pad), `onAction()`, `onTap(x, y)` (canvas pixels mapped to the 240 × 160 grid), `cleanup()`, `msgTop`.
- State the host reads: `won`, `score`, `fails`, `task`. `startMini` routes all input to the object; `endMini(ok, score)` restores play and calls `onEnd`.
- Skip rule: a Skip button appears when `task.attempts ⩾ 2` or `fails ⩾ 2`, and skipping counts as a win. No challenge can block a letter.

Room	Challenge	Mode	Core mechanic and tuning	Fail-safe
0	Royal search	world	3 items shuffled into furniture that has a walkable neighbour	gold sparkle on the next item after 2 misses
1	Fireflies	world + overlay	5 fireflies random-walk at 9 px/s on walkable tiles; catch ring shrinks 34 → 3 px, green zone 7 to 14 px	misses are free and unlimited
2	Rhythm	overlay	33 notes from the theme melody, 84 BPM, one 8-bar pass, judged on the audio clock	always passes, graded
3	Royal recipe	overlay	repeat-the-order memory, rounds of 3, 4 and 5, 0.62 s per step	wrong order replays the round
4	Impress Shadow	overlay	turn-based meter 100 → 0; move powers 22, 32, 20, 28	no loss state
5	Fishing	overlay	bite after 1.4 to 3.6 s, 0.85 s window, 3 catches	early or late just recasts
6	Catch treasures	overlay	12 catches; spawn every max(0.45, 0.95 − 0.012t) s; bolts stun 0.8 s	no loss state
7	Bop-a-Mouse	overlay	12 bops; up-time max(0.65, 1.1 − 0.03b) s	no loss state
8	Frozen cave	world	ice sliding puzzle, 7-move optimum	Shadow fetches the globe after 2 asks
9	Constellations	overlay	2 figures × 8 links, directional star jumping	wrong star just bumps
10	Café	world	4 orders, fetch from counter, carry one item	wrong item gets a joke, retry
11	Lanterns	world	light 4 lanterns to reveal the last letter	none needed

Rhythm chart generation. The chart is derived from the theme, never authored. Lane i follows melodic contour: up moves right, down moves left, big leaps move two lanes, and repeated notes jump across the pad.

```
\ell_i=\operatorname{clamp}\!\big(\ell_{i-1}+\operatorname{sgn}(m_i-m_{i-1})\,(1+[\,|m_i-m_{i-1}|>3\,]),\;0,\;3\big)
```

Hits are judged against `AudioContext.currentTime`, the same clock that schedules the backing track, so audio and judgement cannot drift. With error e = |t_note − t_now|, a hit is Perfect under 80 ms and Good under 170 ms. A hit plays the melody note itself, so the player performs the lead.

Shadow's battle. Each move lowers the Unimpressed meter M by its power p. The treat has a 40% chance to whiff to 6, and repeating the previous move scales it by 0.4, which makes variety the optimal strategy:

```
M_{t+1}=\max\!\big(0,\;M_t-p\cdot(0.4)^{[\,a_t=a_{t-1}\,]}\big)
```

Directional star selection. Pressing a direction picks the star that minimises distance weighted by misalignment, among stars within about 70° of that direction:

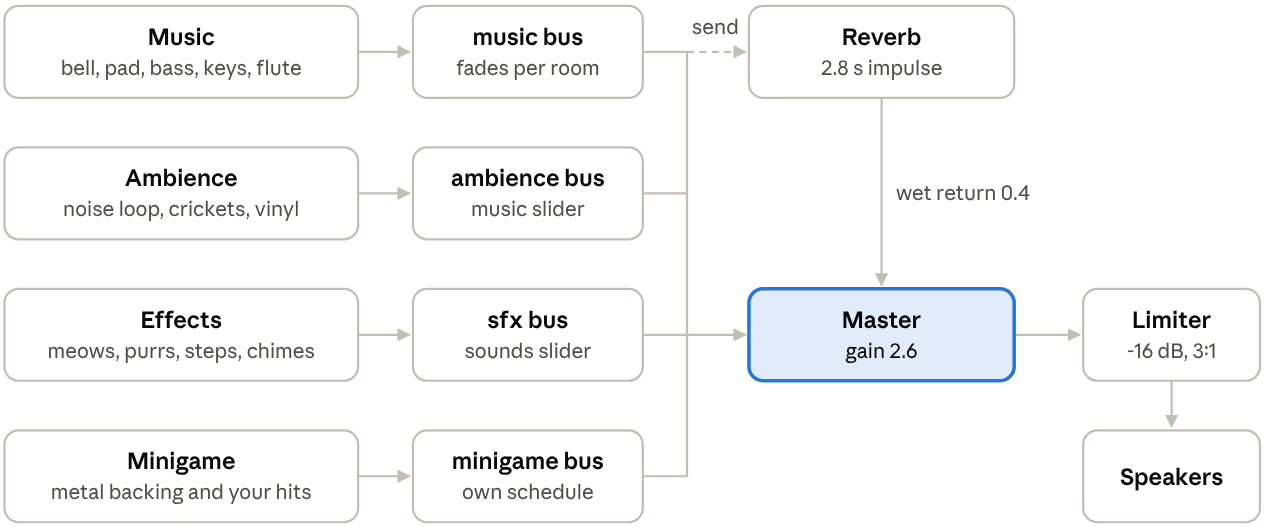
```
j^*=\arg\min_{j:\,\cos\theta_j\ge 0.35}\; d_j\,(2-\cos\theta_j)
```

Ice puzzle, proven solvable. The cave map was generated by random search and accepted only if a breadth-first search over slide states found an optimum of 7 to 9 moves and every reachable state could return to the start. The shipped map has 46 reachable states, a 7-move solution to the snow globe, and no dead ends.

7. Procedural audio engine

The game ships zero audio files: 8 arrangements, 11 instruments and every sound effect are synthesised live with the Web Audio API from one A-minor theme.

Four buses feed one reverb and one limiter, so nothing clips



audio signal graph · 4 buses, 1 reverb, 1 limiter

Sends from the music, effects and minigame buses feed a convolution reverb whose impulse is generated at boot: 2.8 s of stereo white noise under a power-law decay (exponent 2.6). The limiter lets the master run hot at 2.6× without clipping.

Instruments are small functions that build a node chain, schedule an envelope and self-destruct.

Voice	Construction	Used for
Bell	sine + octave partial + 19-semitone triangle ping	melody, chimes, pickups
Pad	two sawtooths detuned ±7 cents per note, 950 Hz low-pass	chord beds
Pluck	triangle through a low-pass swept 4.2 kHz to 600 Hz	harp arpeggios
Keys	sine with 4.5 Hz tremolo and a 4.01× tine partial	lo-fi and rain rooms
Chug, lead	sawtooth fifths and a vibrato square through a tanh(6x) waveshaper	the metal remix
Kick, snare, hat	110 to 40 Hz sweep, band-passed noise with a 190 Hz body, 7 kHz high-passed noise	beats

Theme and arrangement. The theme is 8 bars (Am, F, C, G, Am, F, E, Am) with a 25-note melody. Each room re-voices it at its own tempo from 52 to 84 BPM, and every second loop swaps the melody for a variation, so the same tune never plays identically twice in a row.

Scheduler. A 50 ms timer schedules whole bars whose start falls within 350 ms of the audio clock, the standard lookahead pattern that keeps timing sample-accurate while the main thread renders. Switching style fades the bus out over 0.4 s, resets to bar 1, then fades in over 1.5 s. A guard re-anchors the clock if a background tab falls more than a second behind.

Sound design by synthesis

- Meows: a sawtooth whose pitch rises then falls, through a band-pass formant that sweeps with it. Each cat has a pitch and length signature, from Salem at 1.55× and 0.32 s to Shadow at 0.7× and 0.62 s.
- Purr: low-passed noise amplitude-modulated at 23 Hz, or 27 Hz for Salem.
- Footsteps: noise filtered per floor material, so wood, grass, stone, snow and wet concrete all sound different.
- Letter mode: the music drops to a sparse piano arrangement at 52 BPM, then restores the room's style, or the jukebox choice, on close.

Measured levels. Headless metering on the master bus found the first metal remix peaking at 0.93 of full scale. The chug, kick and snare were cut by about a third, bringing the peak to 0.60 at an RMS of 0.052, in line with the calmer rooms.

8. Persistence

Save data is one JSON document in `localStorage` under `every-version-of-me:v2`, split into a resumable run and an all-time collection that New game never erases.

Key	Shape	Scope	Written when
`run`	`{room, taken[], hearts, badges[], bruce, bruceTalks, bruceSeen[]}` or null	current playthrough	every room entry, heart, badge and Bruce sighting
`notesRead`	letter indices	collection	a letter opens for the first time
`traits`	heart indices	collection	a heart is picked up
`badges`	boolean per room	collection	a host's closing dialogue
`stickers`	sticker ids	collection	a sticker condition fires
`bruce`, `bruceSeen`	flag, room indices	collection	Bruce is found
`outfit`	`{head, clip, neck}`	preference	a badge unlocks an item or the wardrobe changes
`tried`, `wishes`	ids, count	collection	outfit worn, wish made
`settings`	`{music, sfx, textSize, textSpeed}`	preference	the journal settings change
`completed`	boolean	collection	the end card shows; `run` is cleared at the same moment

Checkpoints. The run is checkpointed at room entry, so Continue always restarts the current room from its beginning. Resuming rebuilds every room from source, re-marks collected hearts by `room:x,y` key, and re-attaches Salem as a follower only for rooms 2 to 10.

Forward compatibility. Loaded data is merged over defaults with `Object.assign`, so a save written before a feature existed simply gains that feature's defaults. The key suffix was bumped from `v1` to `v2` when the room count doubled, because room indices changed meaning.

Failure isolation

- Every read and write sits in `try/catch`. If storage is denied, as in some private windows, the game stays fully playable and only Continue disappears.
- The keepsake photo (a PNG data URL) is written under its own key, so a quota failure on the image cannot corrupt the main save.

9. Input and platform

One input layer serves keyboard, mouse and touch, and every path ends in the same three calls: a held direction, an edge-triggered direction, and `pressAction()`.

Keyboard

- Arrows and WASD feed a held set for movement and, in minigames only, an edge-triggered `onDir` that ignores key repeat, so a held arrow never double-hits a rhythm note.
- E, Space, Enter and Z route through `pressAction()`, which dispatches on mode: start, skip intro, advance dialogue, close a letter, act in a minigame, or skip the credits.
- J toggles the journal. Esc closes the journal, a reread letter or a minigame, in that priority.
- Window blur clears every held key, so alt-tabbing never leaves Ellie walking.

Focus arbitration. Enter has two owners: the game and a focused button. Outside play and minigames, Enter and Space fall through to the focused button's native click. Inside play, HUD buttons blur themselves after a click, so Enter always means talk.

Touch

- Touch is detected with `(pointer: coarse)` or `ontouchstart`, which swaps in a 150 px virtual pad and an 86 px action button.

- The pad uses pointer capture and a 14 px dead zone, then picks the dominant axis. Sliding a thumb to a new direction re-fires `onDir`, so the rhythm and recipe games work without lifting.
- Canvas taps are mapped into the 240 × 160 grid for tap-driven minigames, and tapping the dialogue box advances it.

Movement model. Movement is grid-locked at 5.5 tiles per second with linear tweening, and 1.8× that while sliding on ice. The follower steps into the tile the player just vacated, using the same tween, so it never needs its own pathfinding.

Layout. The stage scales to the largest fit in the viewport, reserving 200 px below it in portrait on touch devices for the controls. The page declares `viewport-fit=cover` and pads by the safe-area insets, so notches and home bars never cover the game.

Browser policy. Browsers block audio until a user gesture, so the `AudioContext` is created on the Start or Continue press and resumed if the browser suspended it.

10. Build and distribution

One source file is the release: a single HTML page that loads its two typefaces from Google Fonts and runs properly on her MacBook as it is. The offline USB build is retired.

Release checklist

1. The release is the single file `ellies-journey.html`. There is no build step.
2. Pixelify Sans and IM Fell English load from Google Fonts, each with a system fallback, so the page still works if a font fails to load.
3. Every change passes the static gates and headless runs in section 11 before it is republished.
4. Hand-over check: one full playthrough on her MacBook, in the browser she uses.

Retired: the USB build. It inlined all five font faces as base64 (378,792 bytes) so the game could run with no internet. It was dropped because the HTML page already runs properly on her MacBook, which halves the file and removes a second build to keep in sync.

Companion file. `Every Version of Me - drafts.txt` is generated from the same source of truth: every letter and heart is read out of `CONFIG`, so its final versions cannot drift from the game's.

Hosted build. The hosted artifact has been republished in place at the same link through 16 versions, so a shared link always shows the latest build.

Why not a bundler. There is nothing to bundle. A single hand-written file with no imports opens with a double-click, survives being emailed or copied anywhere, and still runs years from now without a toolchain.

11. Verification

Every version was proven by machines before a person saw it: static gates on every edit, then scripted bots that play each challenge in a real headless browser.

Static gates, on every edit

- Syntax: the script is extracted from the HTML and parsed with `node --check`. Nothing ships if it fails.
- Guarded patches: every code change is a find-and-replace that asserts its anchor matches exactly once, so an edit can never land in the wrong place or silently miss.
- Content lint: every room map is asserted to be 10 rows of 15 characters, and every sprite grid to have equal-width rows.

Runtime tests in headless Chromium

Test	Method	Result
Rhythm scoring	bot hits each note within 40 ms on the audio clock	58 of 58 on the two-pass chart; grade path verified on the 33-note chart
Recipe, fishing, bop, stars	bots read internal state through `_dbg` accessors and answer correctly	every game reaches its win message and the host's closing line
Ice physics	walk onto ice from the entrance	slides 2 tiles and stops against the rock, matching the solver
Ice solvability	breadth-first search over slide states	7-move optimum, 46 states, every state can return to the start
Café orders	fetch and deliver all 4 items	4 of 4 served, task complete, closing dialogue fires
Save and continue	play, reload, press Continue	resumes room 2 with 1 heart and Salem following

Test	Method	Result
Ending chain	light lanterns, read last letter	photo captured, credits roll, post-credits scene, end buttons
Page boot	fresh load in headless Chromium, `file://` URL	0 page errors, audio running; confirmed running on her MacBook
Audio levels	`AnalyserNode` on the master bus	metal remix peak cut from 0.93 to 0.60

Visual review. Each pass produced a screenshot grid of every changed room and overlay, checked by eye before publishing. This caught a wake-up line appearing over the title screen, a letter that opened already scrolled, two ending bubbles overlapping, a missing function that froze the render loop, and a moon that rendered as a square.

What machines cannot check. Whether a letter lands, whether a joke is funny, and whether a challenge feels fair to someone who does not play games. That needs one blind human playthrough before hand-over.

12. Performance and accessibility

The engine's cost is fixed by design: the game always renders 38,400 logical pixels, whatever the screen, and lets the GPU upscale them.

Performance budget

- Framebuffer: 240 × 160 regardless of display size, so a 4K monitor costs the same as a phone. Upscaling is a CSS transform the browser hands to the GPU.
- Per frame: 150 tiles, a few dozen entities and the live particle pool. Sprites are pre-rasterised at boot, so the hot loop is `fillRect` and `drawImage` only.
- Particles live in one typed array filtered each frame. The heaviest emitter, rain, spawns about 90 per second with lifetimes under 1.1 s, which bounds the pool near 100.
- Audio nodes are one-shot: every oscillator and buffer source is scheduled with `stop()`, so the graph garbage-collects itself.
- Speech bubbles are re-anchored per frame only while at least one exists.
- Page size: 189,678 bytes, plus two web fonts fetched from Google Fonts.

Accessibility and comfort

- Reading: letters open in a full-viewport overlay at 17 to 21 px serif, and the Large text setting raises them to 20 to 25 px.
- Pace: text speed has Normal, Fast and Instant settings.
- Sound: separate music and sound-effect sliders, plus a one-tap music toggle in the HUD.
- Motion: `prefers-reduced-motion` turns off bobbing prompts, bubble pops and the timer shake.
- Difficulty: no challenge can be lost, hints appear after misses, and every minigame offers Skip after two failures.
- Contrast: near-black sprites get a light rim so they stay visible on dark floors.
- Semantics: dialogue and minigame messages are `aria-live` regions, letters and the journal are labelled dialogs, and HUD buttons and the photo carry labels.
- Input: every feature works by keyboard alone and by touch alone.

Known gaps. The world itself is a canvas, so a screen reader cannot walk the rooms. The rhythm lanes are colour-coded, although their arrow shapes also carry direction.

13. Architecture decision records

Ten decisions shaped the engine; all are accepted, and each names the condition that should reopen it.

ADR	Decision	Rejected alternatives	Consequence	Revisit when
001	One self-contained HTML file	Godot or Unity export, a zip of assets, a hosted web app	Runs by double-click on any machine; the cost is one long file	the source passes about 5,000 lines
002	Sprites compiled from string grids	PNG sprite sheets, generated images	Zero image assets, and a recolour is one line (Bruce's flame point was); art is capped at 16 px detail	hand-drawn art is wanted

ADR	Decision	Rejected alternatives	Consequence	Revisit when
003	All audio synthesised live	a recorded soundtrack	0 bytes of audio, per-room arrangements, and rhythm judged on the same clock that plays the music; timbre is limited	a recorded original track exists
004	Immediate-mode canvas, no scene graph	DOM sprites, WebGL	Trivially correct redraws at a fixed cost; no draw batching, which 150 tiles never needs	rooms grow past a few thousand tiles
005	One global mode flag, dialogue as an overlay	a state-machine library	Readable dispatch, and cats can talk mid-task; invariants are held by convention, not types	a mode needs to stack on another
006	All content in one `CONFIG` object	JSON files, a CMS	A non-engineer edits one block; the drafts file is generated from it; there is no schema check at load	content is edited by more than one person
007	Extend by rebinding functions, not editing them	refactoring each pass into the core	14 feature passes landed with no regressions; the call chain behind `drawObj` gets deeper each time	before the next major expansion

ADR	Decision	Rejected alternatives	Consequence	Revisit when
008	Guarded patches, one exact anchor each	regenerating the whole file	Every change is small, reviewable and fails loudly on a miss; patches are sensitive to formatting	the file moves to real version control
009	`localStorage` with a run and collection split	IndexedDB, exported save files	Synchronous, tiny and fault-tolerant; saves are per browser and some `file://` contexts deny storage	cross-device saves are wanted
010	No fail states, Skip after two failures	difficulty modes, lives	A letter can never be locked behind skill; the challenge ceiling is deliberately low	the game is ever shown to strangers

14. Known limitations and the overkill roadmap

The shipped engine is deliberately the simplest thing that delivers the letters; this section lists what it cannot do today and the frontier techniques that would take it further.

Known limitations

- The world is a canvas, so it is invisible to screen readers.
- Saves live in one browser on one device, and some `file://` contexts deny storage.
- When the game is played from the hosted link, the photo's Save button is inert; right-click or press-and-hold saving still works.
- No gamepad support, and no remappable keys.
- Debug accessors (`_dbg` , `_rad` , `_notes`) used by the test bots ship in the build. They are read-only and harmless.
- Rebinding chains behind `drawObj` and `drawWall` are up to 3 deep and should be folded back into the core before any further expansion.

Overkill roadmap. Baseline: an immediate-mode Canvas 2D engine with a global mode flag, string-grid sprites, a Web Audio lookahead scheduler and `localStorage` saves. Each option below goes past that, scored from 1 (a library swap) to 10 (specialist-only).

Approach	Complexity	What it adds over the baseline	Pays off when
Seeded PRNG plus input-replay recorder	4/10	Swap `Math.random` for a seeded xorshift and record inputs per frame, so any playthrough replays bit-for-bit as a golden-master regression test	content edits need proof that nothing downstream changed
Entity-component-system with archetype storage	4/10	Behaviours (pettable, follower, speaker, hidden) become composable components instead of per-object conventions	entity count or behaviour combinations grow tenfold
WebGPU renderer with palette-indexed textures	6/10	Real point lights for lanterns and fireflies, palette cycling, a CRT or bloom post-pass, all in one draw call per layer	the art moves past 16 px tiles or wants dynamic lighting
AudioWorklet engine with physical-modelling voices	6/10	Karplus–Strong plucks and modal-synthesis bells rendered sample-accurately off the main thread, plus offline bounce to WAV for a trailer	the soundtrack becomes a feature in its own right

Approach	Complexity	What it adds over the baseline	Pays off when
Wave-function-collapse rooms with solver-gated puzzles	7/10	Rooms generated from tile adjacency rules, with every ice field breadth-first-search proven solvable before it is shown	an endless replay mode is wanted
TLA+ model of progression, model-checked	8/10	A machine proof that every letter is reachable from every save state and no mode transition can deadlock	the story branches or gains optional paths